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2015

数据驱动安全

2015 中国互联网安全大会
China Internet Security Conference

网络空间安全战略论坛



中国互联网安全大会



360互联网安全中心

全球网络空间博弈新常态下的应对策略

沈逸 副教授

复旦大学网络空间治理研究中心 副主任

前引案例

案例（一）：IANA监管权转让

- “考虑转让监管权限” 出台及其影响
- “考虑转让监管权限” 的竞争性解决方案
 - 纯私有化方案
 - 政府间国际组织接管方案
 - 混合方案
- 博弈新常态
 - 从“历史性单一霸权主导” 到梯次修正解决

案例（二）：“监控”与“窃密”

- **APT1、“棱镜门”与OPM**
 - 围绕最具象征意义的核心国家展开的行为规则之争
- **国别行为规范的阐释、解释与适用**
 - “梯队” vs. APT1、“棱镜门”与OPM
- **博弈新常态**
 - 作为“大国专利”的网络安全行为规则博弈
 - 能力的重分配与主权国家的（有限）回归

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- **中美成果清单与GGE报告**
 - 大国网络安全博弈的限度与联合国的作用
- **国家间行为准则的形成与进化**
 - “危机—谈判” 模式
 - 多边模式的有限回归
- **博弈新常态**
 - 混合维度博弈的兴起

内容概要

背景

技术、制度、战略与行为的互动变化

焦点

核心资源、关键话语、主导规则

界定

现实、标准、认知

策略

主导、追随、制衡

一、背景：技术变迁

- 全球范围能力的重新分配与“权力-技术-资本”的互动
- “云端集团”与关键设施供给的“寡头化”
- “终端联盟”与动员能力分布的“多中心化”
- “数据中心”与处理能力的“再中心化”
- “权力中心”、“技术中心”与“资本-市场力量”的互动

销售收入数据展示的欧美“云端集团”的显著优势

Worldwide: Server Vendor Revenue Estimates, 4Q14 (US Dollars)

Company	4Q14 Revenue	4Q14 Market Share (%)	4Q13 Revenue	4Q13 Market Share (%)	4Q14-4Q13 Growth (%)
HP	3,897,668,666	27.9	3,839,527,072	28.1	1.5
Dell	2,425,147,102	17.3	2,074,167,350	15.2	16.9
IBM	1,788,907,117	12.8	3,662,900,853	26.5	-50.6
Lenovo	1,103,620,000	7.9	130,855,009	1.0	743.4
Cisco	772,200,000	5.5	646,100,000	4.7	19.5
Others	3,992,675,197	28.6	3,360,032,355	24.6	18.8
Total	13,980,218,081	100.0	13,673,582,640	100.0	2.2

Note: Lenovo's Q414 results include IBM x86 sales. IBM's Q414 results include only non x86 servers.

Source: Gartner (March 2015)

销售收入数据展示的欧美“数据中心”与处理能力的“再中心化”

Vendor	4Q14	4Q14 Market	4Q13	4Q13 Market	4Q14/4Q13
	Revenue	Share	Revenue	Share	Revenue Growth
1. EMC	\$2,352	22.2%	\$2,276	23.1%	3.3%
2. HP	\$1,456	13.8%	\$1,389	14.1%	4.8%
3. Dell*	\$952	9.0%	\$905	9.2%	5.2%
3. IBM*†	\$951	9.0%	\$1,248	12.7%	-23.8%
5. NetApp	\$764	7.2%	\$791	8.0%	-3.5%
ODM Direct	\$1,357	12.8%	\$973	9.9%	39.4%
Others	\$2,740	25.9%	\$2,281	23.1%	20.2%
All Vendors	\$10,571	100.0%	\$9,864	100.0%	7.2%

“权力中心”、“技术中心”与“资本-市场力量”的互动

siliconbeat

The tech blog of The Mercury News | Home to Good Morning Silicon Valley

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Social Web Mobile Cloud Gadgets Cleantech VC Data Games Policy Media

Quoted: Microsoft vs. U.S. government on email privacy

By Levi Sumagaysay / September 8, 2015 at 9:55 AM



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@petecarey

2m

iPhone impress consumers, not Wall Street:
bayareane.ws/1QJhF7J

Show Summary

Obama Hosts CyberSecurity Summit But Google, Yahoo, and Facebook Show Him Exactly How Much They Care

BY MICHAEL HAUSAM (7 MONTHS AGO) | NATION, POLITICS, TECHNOLOGY



Getty - Nicholas Kamm

What if you threw an internet security party and the planet's three biggest companies refused to attend? That's what happened to the President on Friday at Stanford University.

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Dear God,
I want to be just
like my Daddy when I get
big but not with so much

15 Hilariously INSENSITIVE
Notes From Kids



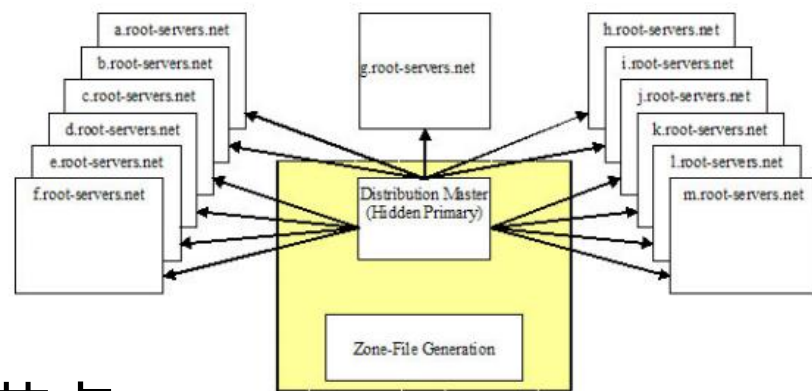
一、背景：概念变化

- **组织观念的变化与多利益相关方的变迁**
 - 多利益相关方组织的独立性、自主性与问责
- **主权观念的变化与管辖的争议**
 - 安全焦虑与主权管辖的回归
 - 霸权焦虑与主权管辖的扩展
 - 隐私焦虑与主权管辖的限制和让渡

二、焦点：核心资源

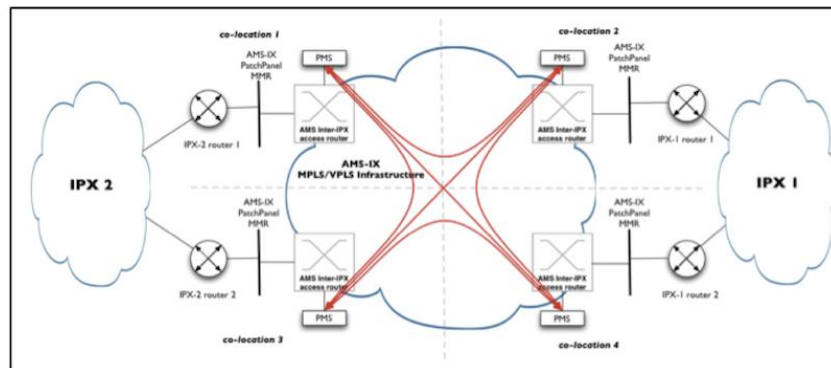
• 全球层面的核心资源

- 根服务器及其镜像
- 隐藏发布主机
- 骨干通讯链路与国际枢纽节点
- 全球性数据资源中心



• 国家层面的核心资源

- 国际出口与国内通讯链路
- 等级地理域名解析服务器
- 国家级数据资源中心



1¢ 14¢
FIXED-LINE MOBILE

Per-minute wholesale cost to send an international call to Switzerland.

27%

TRAFFIC SENT AS VoIP
Share of total international voice traffic transported as VoIP by carriers in 2009.

4/5

WIRELESS WORLD
Share of phones worldwide that are mobile.

12%

SKYPE VS. THE WORLD
Total international Skype traffic as a percentage of world-wide international voice traffic in 2009, approximately 33 billion minutes.0.1 4.1
FIXED-LINE MOBILE

Billions of new subscribers added since 2000.

GLOBAL TRAFFIC MAP 2010

TeleGeography ROGERS



MINUTES & VOLUME



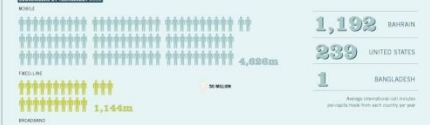
PRICING & REVENUES



USERS & TECHNOLOGY



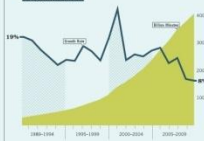
TECHNOLOGY & INNOVATION



INTERNATIONAL TRAFFIC FLOWS 2009



TRAFFIC TRENDS & GROWTH



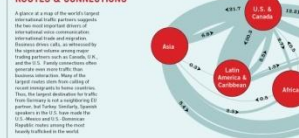
TRAFFIC TRENDS & GROWTH



TRAFFIC TRENDS & GROWTH



ROUTES & CONNECTIONS



INTERNATIONAL TRAFFIC FLOWS 2009



二、焦点：关键话语

- 全球网络空间
- 国家安全
- 用户权益与个人隐私
- 行为规范



二、焦点：主导规则

- 技术-产业-行业标准
- 国家行为边界判定标准
- 全球网络空间主导规则



三、界定：现实

- 从历史性单一霸权向多中心体系的变迁
- 渐进-可逆的国家-个体行为边界博弈



三、界定：标准

- “国际”/社群标准
- 国家标准
- 个体标准



NTIA IANA Functions' Stewardship Transition

OVERVIEW & TIMELINE COORDINATION GROUP TRANSITION RESOURCES COMMUNITY DISCUSSIONS EVENTS CALENDAR

This page is available in: English | العربية | Español | Français | Русский | 中文

Overview

Welcome to the NTIA IANA Functions' Stewardship Transition microsite, a comprehensive destination for discussion, contribution, and collaboration related to the NTIA's announcement and request of 14 March 2014.

As convener of the process to develop a transition proposal, ICANN has made available the tools and resources to facilitate the discussion and capture the views and comments of the global multistakeholder community.

This microsite will serve as a resource and information-sharing platform for all interested parties, as well as a tool for the IANA Stewardship Transition Coordination Group (ICG) to keep global stakeholders updated on all developments.

Quick Links

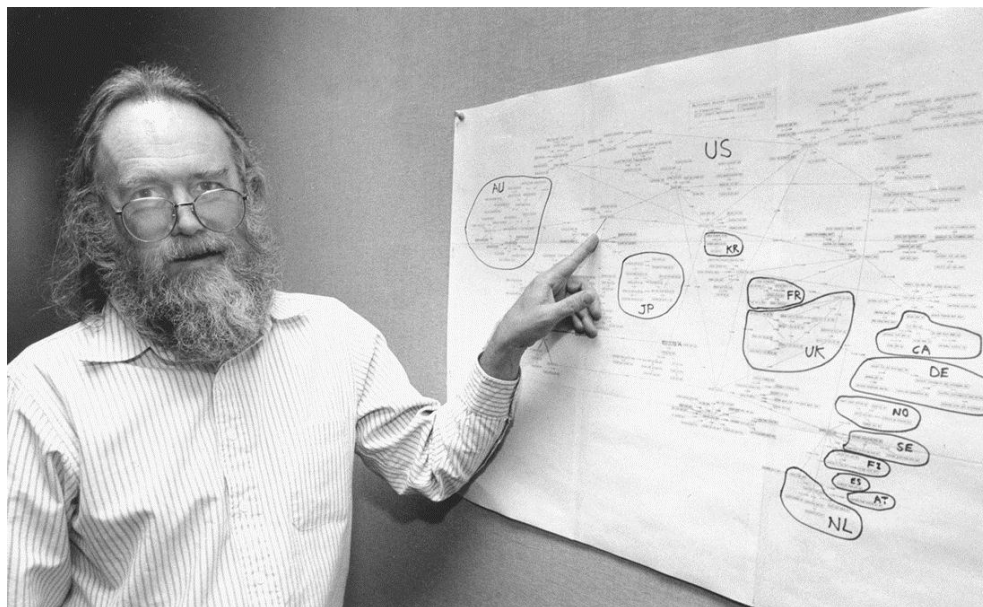
- [ICG Website](#)
- [Members](#)
- [Documents and Calls for Input](#)
- [Meetings and Archives](#)
- [Communications](#)

Background

Shortly after the [NTIA Announcement](#) of 14 March 2014, ICANN launched a multistakeholder process and discussion to gather community views and input on the principles and mechanisms for the transitioning of NTIA's stewardship of the IANA functions.

Recent News and Announcements

ICG – Sixth Face-to-Face Meeting in Los Angeles



三、界定：认知

- 认知相符与国家思维定势
- 威胁认知与战略选择偏好

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ASIA DEFENSE

China-US War Over Taiwan and Spratlys: Which Side Would Win?

A new RAND report analyzes US and Chinese military capabilities based on two future war scenarios.

By Franz-Stefan Gady
September 28, 2015

0 Likes
0 Comments

While the People's Liberation Army (PLAN) is still far away from catching up with U.S. military aggregate capabilities, Chinese forces can still pose a formidable challenge to American power in China's immediate periphery.

That's the major conclusion of a new 430-page **RAND** report written by 14 scholars titled "The U.S.-China Military Scorecard: Forces, Geography, and the Evolving Balance of Power, 1996-2017."

Image Credit: Wikimedia Commons/ Jian Kang



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Cruz foiled in bid for
Planned Parenthood...
07:04pm

HEALTHCARE

Cruz foiled in bid for
Planned Parenthood...
07:04pm

FLOO

Cruz: Boel
lost his job
07:36pm

HOME | POLICY | CYBERSECURITY | OVERNIGHT CYBERSECURITY

Overnight Cybersecurity: Emerging cyber war rules spark debate

f s+ t 33 in 7

COMMENTS 0

By Cory Bennett and Katie Bo Williams - 09/21/15 07:14 PM EDT

Welcome to **OVERNIGHT CYBERSECURITY**, your daily rundown of the biggest news in the world of hacking and data privacy. We're here to connect the dots as leaders in government, policy and industry try to counter the rise in cyber threats. What lies ahead for Congress, the administration and the latest company under siege? Whether you're a consumer, a techie or a D.C. lifer, we're here to give you ...

THE BIG STORIES:

--I'M KIND OF A BIG DEAL: Experts disagree wildly on the scope and utility of a possible agreement between the United States and China promising that neither country will be the first to launch cyberattacks on the other's critical infrastructure. Although President Obama and President Xi Jinping are not expected to reveal any specifics of the accord during the latter's state visit this week, they will likely announce a "generic embrace" of a code of conduct recently adopted by the United Nations that includes a tenet prohibiting states from engaging in cyber activity that intentionally damages critical infrastructure. Because the new accord likely will not address the more high-profile forms of hacking China is accused of carrying out in the U.S., like the theft of intellectual property, some experts have criticized it as too narrow in scope. Paul Stockton, the assistant secretary of Defense for homeland defense from 2009 to 2013, called the proposed agreement "constructive" but "very limited in terms of its intrinsic value." Others suggest it has broader international implications. Atlantic Council senior fellow Jason Healey says embracing the UN guidelines could put pressure on other, more dangerous nations -- like Russia -- to sign the week, Xi Jinping is not going to take down electronic U.S. infrastructure."

四、策略：主导

- 从历史性霸权到自觉主导
 - 互联网发展的历史惯性客观上形成了单一主权国家主导网络关键资源的事实
 - 这个事实与互联网技术的内生特点之间存在微妙的关系
 - 自觉强化主导和塑造能力，谋求网络空间的行动自由
- 主导策略的成本与收益

建设、维持和巩固霸权优势（1）

THE DEPARTMENT OF DEFENSE CYBER STRATEGY

The purpose of this strategy is to guide the development of DoD's cyber forces and strengthen our cyber defense and cyber deterrence posture. It focuses on building cyber capabilities and organizations for DoD's three primary cyber missions.

DoD's Three Primary Cyber Missions:



Defend DoD networks, systems, and information



Defend the U.S. homeland and U.S. national interests against cyberattacks of significant consequence



Provide cyber support to military operational and contingency plans

Cyber Mission Force: 133 teams by 2018

State and non-state actors threaten disruptive and destructive attacks against the United States and conduct cyber-enabled theft of intellectual property to undercut the United States' technological and military advantage. DoD must develop its cyber forces and strengthen its cyber defense and cyber

THE DOD CYBER STRATEGY



THE DEPARTMENT OF DEFENSE

[Full Strategy](#) [Fact Sheet](#)

"America's economic prosperity, national security, and our individual liberties depend on our commitment to securing cyberspace and maintaining an open, interoperable, secure, and reliable Internet."

President Barack Obama
Feb. 13, 2015
Full Statement

"We're the ones who stand with those who create and innovate against those who would steal and destroy. That's the kind of country we are, and that's the kind of cyber

建设、维持和巩固霸权优势（2）

Verisign/ICANN Proposal in Response to NTIA Request

Root Zone Administrator Proposal Related to the IANA Functions Stewardship Transition

Introduction

On March 14, 2014, NTIA announced its intent to transition key Internet domain name functions to the global multistakeholder community. As the first step, NTIA asked the Internet Corporation for Assigned Names and Numbers (ICANN) to convene global stakeholders to develop a proposal related to the stewardship of the Internet Assigned Numbers Authority (IANA) function. Just over 17 months later, stakeholders have issued a draft proposal on this as well as a proposal to enhance the accountability of ICANN, with the intention of a final package coming to NTIA later this year.

In order to prepare for the implementation phase of the IANA stewardship transition, NTIA has asked Verisign and ICANN to submit a proposal as to how best remove the NTIA's administrative role associated with root zone management in a manner that maintains the security, stability and resiliency of the Internet's domain name system. This proposal offers a high-level plan for the transitioning of the NTIA role for consideration along with joint recommendations from Verisign and ICANN on the way forward.

Background

The Root Zone Management System (RZMS) is a set of tools, which currently allows ICANN as the IANA Functions Operator (IFO), Verisign, as the Root Zone Maintainer (RZM), and the National Telecommunications and Information Administration (NTIA) at the U.S. Department of Commerce (DoC), as the Root Zone Administrator (RZA), to collaboratively manage the changes to the Internet's single authoritative root zone. The current automated RZMS replaced the legacy root zone update, validation, extraction, and distribution methods with a web based user interface (UI) that permits Top-Level Domain (TLD) managers to submit change requests, an Extensible Provisioning Protocol (EPP) client and server system that communicates incoming root zone change requests between the IFO and the RZM, two web-based authorization UI portals for NTIA use, a set of special RZMS web based UIs for ICANN and Verisign to facilitate administrative management of change requests, customer service representative (CSR) tools to allow IFO and RZM staff to perform functions such as checking domain history and accommodating any required manual updates, and dedicated mail servers for sending and receiving encrypted and signed RZMS-related electronic messages.

Figure 1 illustrates the relationship between the three parties involved in root zone management and their respective functions and roles.

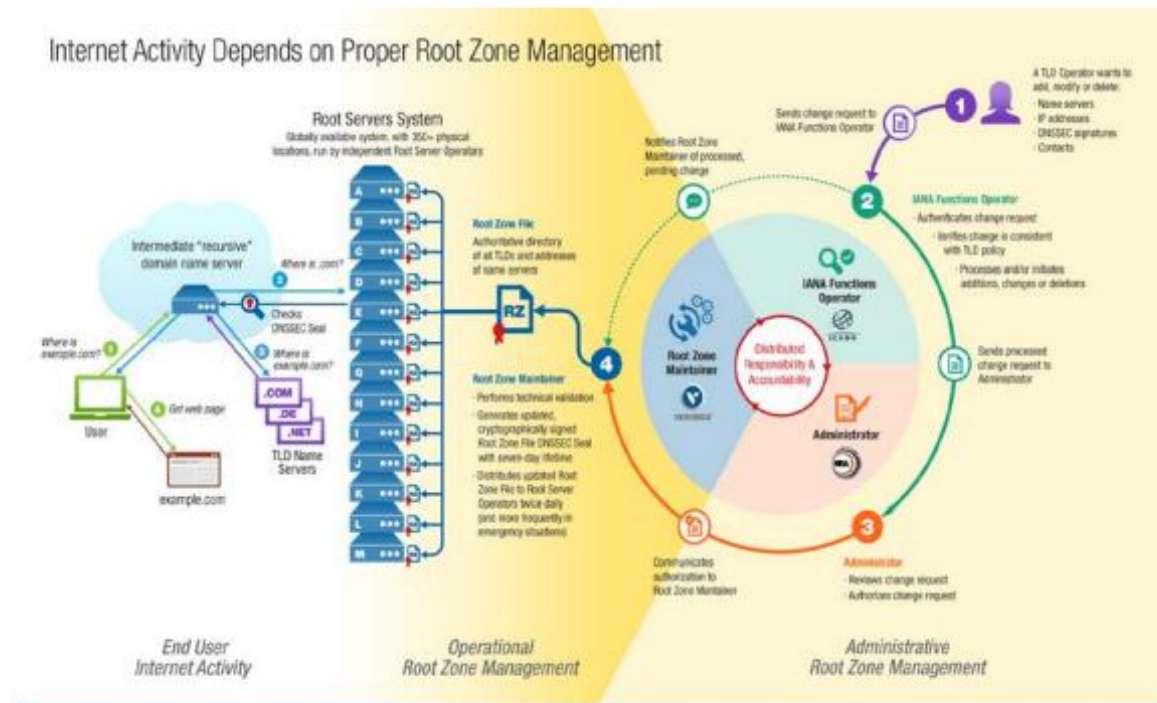


Figure 1: The Root Zone Management Functions

建设、维持和巩固霸权优势（3）

The New York Times | <http://nyti.ms/1Hc1QzG>

ASIA PACIFIC

U.S. Decides to Retaliate Against China's Hacking

点击查看本文中文版 | Read in Chinese

By DAVID E. SANGER JULY 31, 2015

The Obama administration has determined that it must retaliate against China for the theft of the personal information of more than 20 million Americans from the databases of the Office of Personnel Management, but it is still struggling to decide what it can do without prompting an escalating cyberconflict.

The decision came after the administration concluded that the hacking attack was so vast in scope and ambition that the usual practices for dealing with traditional espionage cases did not apply.

But in a series of classified meetings, officials have struggled to choose among options that range from largely symbolic responses — for example, diplomatic protests or the ouster of known Chinese agents in the United States — to more significant actions that some officials fear could lead to an escalation of the hacking conflict between the two countries.

That does not mean a response will happen anytime soon — or be obvious when it does. The White House could determine that the downsides

http://www.nytimes.com/2015/08/01/world/asia/us-decides-to-retaliate-against-chinas-hacking.html?_r=0

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ASIA-PACIFIC NEWS

US to hit China hackers before Xi's Washington visit

Demetri Sevastopulo, Gina Chon and Charles Clover
Thursday, 3 Sep 2015 | 5:21 PM ET

FINANCIAL TIMES



Damir Sagolj | Reuters

Chinese President Xi Jinping waves at the beginning of the military parade marking the 70th anniversary of the end of World War Two, in Beijing, September 3, 2015.

The White House is preparing to slap sanctions as early as next week on Chinese companies connected to the cyber theft of US intellectual property.

四、策略：追随

- 从联盟到追随策略
- 追随策略的成本与收益



Deliberate Deception: Washington Gave Answer Long Ago in NSA Case

SPIEGEL ONLINE - August 21, 2015

For months, the German government sought to create the impression it was still waiting for an answer from the US on whether it could share NSA target lists for spying with a parliamentary investigation. The response came months ago. By Matthias Gebauer, René Pfister and Holger Stark [more...](#) [Comment]



'Secrets Must Remain Secret': German Intelligence Coordinator on NSA and Media Leaks

SPIEGEL ONLINE - August 14, 2015

In a SPIEGEL interview, Chancellor Angela Merkel's Chief of Staff Peter Altmaier speaks of mistakes made by German intelligence, leaks of confidential information in Berlin and German timidity in the face of US spying. Interview Conducted By Michael Sauga, Jörg Schindler and Peter Müller [more...](#) [Comment]



Berlin's NSA Fears: Treason Investigation Reveals Anxiety at the Top

SPIEGEL ONLINE - August 10, 2015

Germany's federal prosecutor was just forced out of office after launching a treason investigation against journalists. The scandal, just like the "SPIEGEL Affair" 50 years ago, reveals a growing state of anxiety among the country's political elite. A Commentary by Klaus Brinkbäumer [more...](#) [Comment]



Japan calls reports of US spying 'deeply regrettable'

WikiLeaks revelations of NSA surveillance of Japanese politicians could strain relationship between key allies

August 3, 2015 4:30PM ET

Japan has described claims that the United States spied on Japanese politicians and major firms as "deeply regrettable" in its first official response to revelations by the whistleblower group WikiLeaks.

The latest WikiLeaks intercepts exposing U.S. National Security Agency (NSA) activities follow other documents that revealed spying on allies including Germany and France.

四、策略：制衡

- 从硬挑战到软制衡
- 从否定性制衡到建设性制衡



23, 24 - April 2014

Grand Hyatt Hotel
São Paulo, Brazil



Report of the Working Group on Internet Governance

Château de Bossey
June 2005

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15. Administration of the root zone files and system

Unilateral control by the United States Government.

- For historical reasons, the existing system involves only one Government in the authorization of changes to the root zone file.

Lack of formal relationship with root server operators.

- The root zone operators perform their functions today without a formal relationship with any authority.

力量对比变迁与新的动态平衡

SEP 24, 2015 @ 03:02 PM 26,318 VIEWS

The \$2.5 Trillion Photo: China's President Xi Jinping Meets With Heads Of Apple, Amazon, And More



Liyan Chen
FORBES STAFF

Charting the world's
largest companies and
wealthiest people.

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FULL BIO >



China's President Xi Jinping with top U.S. and Chinese tech executives in Seattle (Credit: The Associated Press)

A picture is worth a thousand words. In the first official U.S. visit of China's President **Xi Jinping** this week, nothing speaks louder than one single photo on China's hopes for its high-tech diplomacy.

传承历史渊源

塑造未来



拜耳材料科技更名为Covestro (中文译名: 科思创)

中美达成的互联网成果要点

持续展开的动态博弈与新的均衡点：行为规范、边界和模式

6、中美双方承诺，其各自关于外资的国家安全审查(在美方指美国外国投资委员会的审查程序)的范围限于属于国家安全关切的问题，不通过纳入其他更宽泛的公共利益或经济问题将审查范围泛化。

双方承诺，其各自的国家安全审查对于被审查的每项投资，无论其来源地都按照法律适用相同的规则 and 标准。当一项投资存在国家安全风险时，双方将尽可能快地通过各自审查程序解决风险，包括在合理且可能的情况下，采取有针对性的缓和措施，而不是禁止投资。中美两国各自的国家安全审查仅适用于其审查制度建立后开展的投资。当一项投资在中国或美国通过国家安全审查程序后，如投资方按照审查通过的方案完成了投资，则该项投资在通常情况下不应被再次审查。

双方承诺，在各自的国家安全审查中，不将非投资方提供的信息用以与国家安全无关、促进投资方竞争者商业利益的目的。

中美双方承诺，未来就两国国家安全审查的有关问题继续开展交流，包括各自国家安全审查程序的范围、各自国家安全审查程序中非投资方的作用。

11、技术是中美双边经济关系的支柱之一。为更广阔的技术产业双向贸易和投资创造条件并避免限制措施，对保持两国经济关系的积极势头至关重要。

(3)双方承诺，用以在商业领域加强信息通信技术网络安全(信息通信技术网络安全法规)的一般适用措施，应符合世贸组织协议，仅用于小范围，考虑国际规范，非歧视，且不对商业机构在相关产品的购买、销售或使用方面不必要地设置基于国别的条件或限制。

中美双方同意，就恶意网络活动提供信息及协助的请求要及时给予回应。同时，依据各自国家法律和有关国际义务，双方同意就调查网络犯罪、收集电子证据、减少源自其领土的恶意网络行为的请求提供合作。双方还同意适当向对方提供调查现状及结果的最新信息。

中美双方同意，各自国家政府均不得从事或者在知情情况下支持网络窃取知识产权，包括贸易秘密，以及其他机密商业信息，以使其企业或商业行业在竞争中处于有利地位。

双方承诺，共同继续制定和推动国际社会网络空间合适的国家行为准则。中美双方欢迎2015年7月联合国“从国际安全角度看信息和通讯领域的发展”政府专家组报告，该报告旨在处理网络空间的行为准则和其他涉及国际安全的重要问题。双方也同意，就此话题建立一个高级专家小组来继续展开讨论。

中美双方同意，建立两国打击网络犯罪及相关事项高级别联合对话机制。中方将指定一名部级领导牵头，公安部、国家安全部、司法部和国家网信办参加。美国国土安全部部长和司法部部长将作为对话的美方共同团长，由联邦调查局、美国情报委员会和其他部门的代表参加。该机制对任一方关注和发现的恶意网络行为所请求的反馈信息和协助的时效性和质量进行评估。作为机制的一部分，双方同意建立热线，以处理在响应这些请求过程中可能出现的问题升级。最后，双方同意对话第一次会议于2015年内举行，之后每年两次。

路径：传统路径与新路径

- **传统军备控制路径**

实力变迁引发危机，危机促进对话，对话建立信任，形成新的稳定模式；以一定程度的“相互确保毁灭”或者对“相互确保毁灭”的默认，作为战略稳定的基础。优点在于非常成熟的知识与实践，缺点在于巨大的机会成本，以及潜在的风险。

- **新的合作路径**

超越传统军备控制模式，寻找新的建设性的合作模式



中国互联网安全大会



360互联网安全中心

感谢聆听！